PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

The project **“Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking”** is designed to simplify the process of importing CSV data, mapping fields dynamically, and establishing relationships between datasets. It provides a system that handles data processing, validation, and relationship creation without relying on web design elements.

1.2 Purpose

The main purpose of the project is to:

- Reduce manual data entry
- Ensure accurate data mapping
- Automate relationship creation between datasets
- Improve efficiency in data handling

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2. IDEATION PHASE

2.1 Problem Statement

Organizations often deal with large datasets that need to be imported and structured properly. Manual data entry and mapping are time-consuming, error-prone, and inefficient.

2.2 Empathy Map Canvas

- **Users Think:** Data import is complex and confusing
- **Users Feel:** Frustrated with manual work
- **Users Say:** “We need faster data processing”
- **Users Do:** Manually enter or clean data

2.3 Brainstorming

- Automate CSV import and parsing
- Implement dynamic field mapping
- Add validation and error-checking
- Automate relationship mapping between datasets

- Keep interface minimal and functional

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User provides CSV file
2. System reads and parses data
3. User maps fields (logic-based)
4. System validates mapping
5. System processes relationships
6. Structured results are generated

3.2 Solution Requirement

- CSV import functionality
- Data parsing and storage (local or in-memory)
- Field mapping mechanism
- Validation logic for correctness
- Relationship processing system
- Output result generation

3.3 Data Flow Diagram (Text Representation)

User → Provide CSV → Parse Data → Store Temporarily

→ Field Mapping → Validate Mapping → Save Mapping

→ Process Relationships → Generate Result

3.4 Technology Stack

- Core Language: Python / JavaScript (logic-based, non-UI)
- Storage: Local storage or in-memory structures
- Tools: Any script execution environment

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4. PROJECT DESIGN

4.1 Problem-Solution Fit

The project eliminates manual data handling errors and inefficiencies by automating import, mapping, and relationship creation.

4.2 Proposed Solution

- Accept CSV input
- Dynamically map fields
- Process and create dataset relationships
- Produce structured output

4.3 Solution Architecture

Input Layer → CSV Import

Processing Layer → Parsing & Field Mapping

Validation Layer → Data Checks & Relationship Logic

Output Layer → Structured Result

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5. PROJECT PLANNING & SCHEDULING

- **Week 1:** Idea & Requirement Analysis
- **Week 2:** Logic Design & Data Flow
- **Week 3:** CSV Import & Parsing Module
- **Week 4:** Mapping & Validation Logic
- **Week 5:** Relationship Processing Module
- **Week 6:** Testing & Documentation

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Tested with small and large CSV datasets
- Verified mapping accuracy
- Ensured duplicate prevention and relationship correctness
- Validated output structure against expected results

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7. RESULTS

7.1 Output Representation

- **Uploaded Data:** Parsed correctly
- **Field Mapping:** Correctly assigned
- **Processed Relationships:** Established accurately
- **Final Output:** Structured dataset ready for use

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8. ADVANTAGES

- Reduces manual work significantly
- Ensures accurate data mapping
- Prevents errors in relationships
- Fast processing without UI overhead

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9. CONCLUSION

The project successfully automates data import and relationship mapping, improving accuracy and efficiency in handling structured datasets. It is functional without web interface dependencies, making it adaptable for backend or script-based implementations.